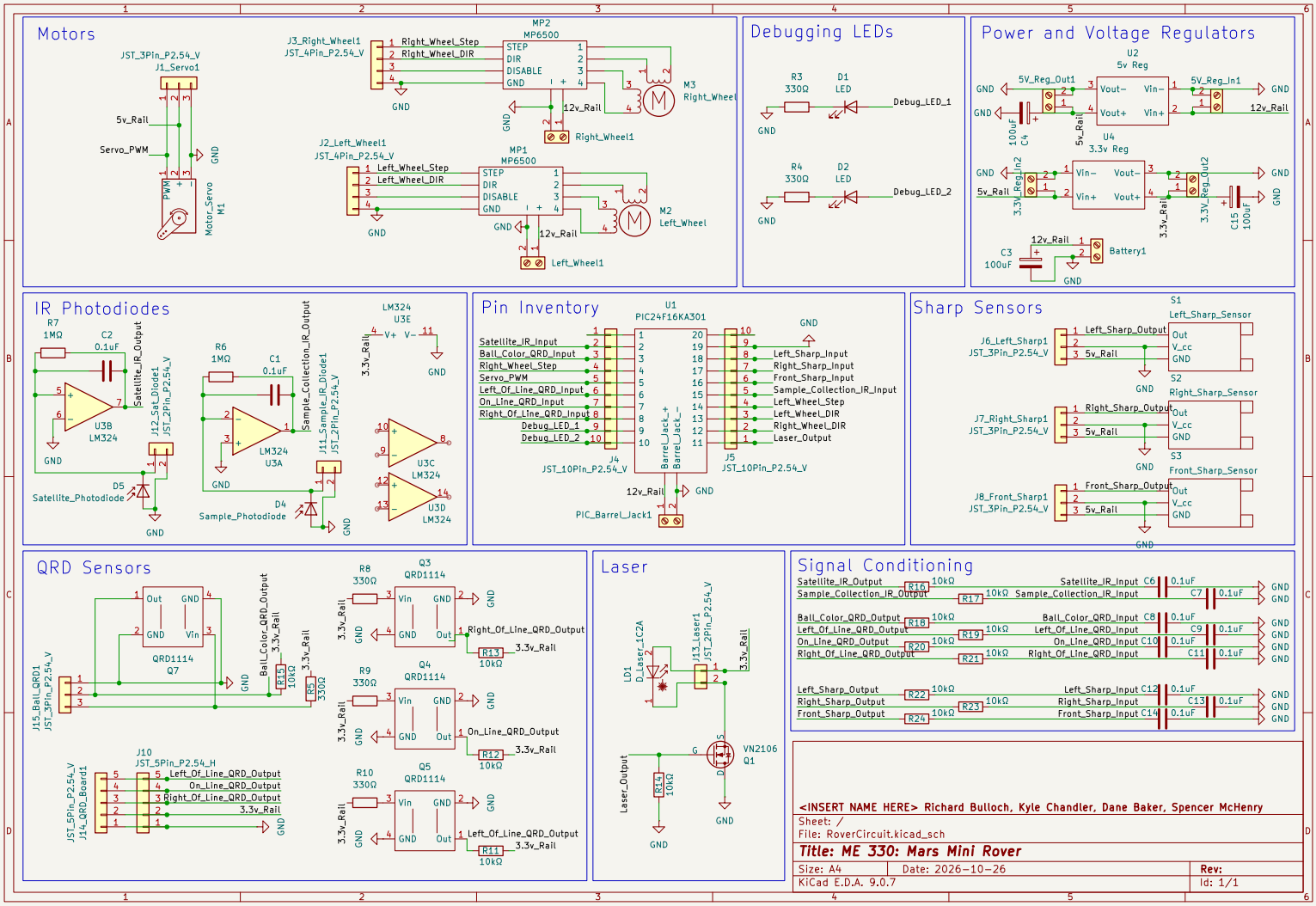

Motors
JST_3Pin_P2.54_V
J1_Servo1
5v_Rail
Servo_PWM
GND
PWM
Motor_Servo
M1
J3_Right_Wheel1
JST_4Pin_P2.54_V
Right_Wheel_Step
Right_Wheel_DIR
MP2
MP6500
STEP
DIR
DISABLE
GND
M3
Right_Wheel
12v_Rail
Right_Wheel1
J2_Left_Wheel1
Left_Wheel_Step
Left_Wheel_DIR
MP1
M2
Left_Wheel
Left_Wheel1
Debugging LEDs
R3
330Ω
D1
LED
Debug_LED_1
R4
D2
Debug_LED_2
Power and Voltage Regulators
U2
5v Reg
5V_Reg_Out1
5V_Reg_In1
Vout−
Vin−
Vout+
Vin+
100uF
C4
5v_Rail
U4
3.3v Reg
3.3V_Reg_In2
3.3V_Reg_Out2
C15
C3
Battery1
IR Photodiodes
R7
1MΩ
C2
0.1uF
Satellite_IR_Output
U3B
LM324
J12_Sat_Diode1
JST_2Pin_P2.54_V
D5
Satellite_Photodiode
R6
C1
Sample_Collection_IR_Output
U3A
J11_Sample_IR_Diode1
D4
Sample_Photodiode
LM324
U3E
3.3v_Rail
U3C
U3D
Pin Inventory
U1
PIC24F16KA301
Satellite_IR_Input
Ball_Color_QRD_Input
Right_Wheel_Step
Servo_PWM
Left_Of_Line_QRD_Input
On_Line_QRD_Input
Right_Of_Line_QRD_Input
Debug_LED_1
Debug_LED_2
Barrel_Jack_+
Barrel_Jack_−
J4
JST_10Pin_P2.54_V
J5
Left_Sharp_Input
Right_Sharp_Input
Front_Sharp_Input
Sample_Collection_IR_Input
Left_Wheel_Step
Left_Wheel_DIR
Right_Wheel_DIR
Laser_Output
PIC_Barrel_Jack1
Sharp Sensors
S1
Left_Sharp_Sensor
J6_Left_Sharp1
JST_3Pin_P2.54_V
Left_Sharp_Output
Out
V_cc
5v_Rail
S2
Right_Sharp_Sensor
J7_Right_Sharp1
Right_Sharp_Output
S3
Front_Sharp_Sensor
J8_Front_Sharp1
Front_Sharp_Output
QRD Sensors
J15_Ball_QRD1
JST_3Pin_P2.54_V
QRD1114
Q7
Ball_Color_QRD_Output
R15
10kΩ
R5
330Ω
R8
330Ω
Q3
R13
Right_Of_Line_QRD_Output
R9
Q4
R12
On_Line_QRD_Output
R10
Q5
R11
Left_Of_Line_QRD_Output
J10
JST_5Pin_P2.54_H
JST_5Pin_P2.54_V
J14_QRD_Board1
Laser
LD1
D_Laser_1C2A
J13_Laser1
Laser_Output
R14
VN2106
Q1
Signal Conditioning
Satellite_IR_Output
R16
Satellite_IR_Input
C6
C7
Sample_Collection_IR_Output
R17
Sample_Collection_IR_Input
Ball_Color_QRD_Output
R18
Ball_Color_QRD_Input
C8
Left_Of_Line_QRD_Output
R19
Left_Of_Line_QRD_Input
C9
On_Line_QRD_Output
R20
On_Line_QRD_Input
C10
Right_Of_Line_QRD_Output
R21
Right_Of_Line_QRD_Input
C11
Left_Sharp_Output
R22
Left_Sharp_Input
C12
Right_Sharp_Output
R23
Right_Sharp_Input
C13
Front_Sharp_Output
R24
Front_Sharp_Input
C14
10kΩ
0.1uF
<INSERT NAME HERE> Richard Bulloch, Kyle Chandler, Dane Baker, Spencer McHenry
Sheet: /
File: RoverCircuit.kicad_sch
Title: ME 330: Mars Mini Rover
Size: A4
Date: 2026-10-26
Rev:
KiCad E.D.A. 9.0.7
Id: 1/1